

LinAge2 Query Battery

52 queries against a patient someone types into the *My Patient* tab — each with the query form it should translate to, the behaviour that counts as a pass, and what the system actually returned on 2026-10-04.

What it is for. A repeatable acceptance pass over the LinAge2 path: plain text in, a patient read back and built for one browser session, the clinical clock scored in-process, questions about that patient translated, routed to the right space and answered. Part A tests the tab; Parts B–D the inference over the clock; Part E questions that cross into the rest of the knowledge base; Part F honesty.

What it is not. Not a benchmark with a score, and not a clinical tool. The interesting signal is where years are reported without a cause, where an empty answer is correct, and where a refusal protects a number from a wrong unit.

Verified, not asserted. 47 of the 52 entries were executed by `scripts/linage2_battery.py` — every query form through `POST /query` with only the LLM translator stubbed (system prompt, patient block, routing, validation, worker-process execution, warnings all as served), every tab entry through the tab's own handlers — and 47 passed. The other 5 have no query form (∅): their correct answer is a direct reply, which needs the LLM translator and is graded live. Translation itself (question → form) is likewise graded live.

Reading an entry

Each entry carries the question as a person would type it, the form the translator should emit ($\emptyset$ means a direct answer and nothing executed), the pass criterion, and OBSERVED: what the system returned when the form was run against the named patient.

THE DISTINCTION THE BATTERY IS BUILT AROUND

Years are not causes. LinAge2 gives every lab a number of years, but its per-lab weight is a sex-specific projection whose sign is not the lab's clinical direction (CRP is +4.4 months per SD in the female model and -0.2 in the male one). So the knowledge base credits a cause only when the person's own value witnesses it — their HbA1c is elevated, they said they smoke. Everything else is reported as years and left unexplained.

Three shapes follow from that and recur below: a lab that adds years with no cause (albumin); a lab that is high and still takes years off (glucose in the male model); and a lever that returns zero because there is nothing witnessed for it to act on.

RUNTIME CONDITIONS THAT CHANGE HOW A RESULT READS

The patient lives in one browser session. It is a plain dict in the UI's session state; every question rebuilds its atoms into that one query's space. Nothing is written to the knowledge base.

LinAge2 runs in its own space. Its atoms never enter the shared space (they crash it); a question that needs both is split, and the answers are joined in the order asked.

An empty result is frequently the pass. No absolute risk without a baseline, no lever without a route, no LinAge2 for the built-in patients.

PATIENTS

Who the questions are about

Five patients, each typed into the tab as plain text and built by the tab's own reader and the in-process LinAge2, always as Caller_Me. P1–P3 are the tab's example buttons, shown below exactly as typed; P4 and P5 are one-line variations of P1.

P1 the 58-year-old smoker example

P2 the healthy 45-year-old woman example

P3 the six-labs example (66, male, former smoker)

P4 P1 plus a line 'GrimAge acceleration +4.5 years'

P5 P1 with 'current smoker' replaced by 'cotinine 250 ng/mL'

P1 · AS TYPED

58 year old male, current
smoker
albumin 4.1 g/dL
HbA1c 6.4 %
CRP 3.1 mg/L
fasting glucose 112 mg/dL
blood pressure 142/88
pulse 78
total cholesterol 228 mg/dL
HDL 41 mg/dL
triglycerides 190 mg/dL
creatinine 1.1 mg/dL
RDW 14.1 %
WBC 8.2
BMI 29.4
diagnoses: hypertension
self-rated health: fair

P2 · AS TYPED

45 year old female, never
smoked
albumin 4.5 g/dL
HbA1c 5.2 %
CRP 0.6 mg/L
fasting glucose 88 mg/dL
blood pressure 112/72
pulse 62
total cholesterol 182 mg/dL
HDL 68 mg/dL
triglycerides 75 mg/dL
creatinine 0.8 mg/dL
RDW 12.6 %
hemoglobin 13.4 g/dL
platelets 245
weight 61 kg
height 168 cm
no known conditions
self-rated health: very good

P3 · AS TYPED

age 66, male, former smoker
albumin 3.8 g/dL
HbA1c 7.1 %
CRP 6.5 mg/L
creatinine 1.3 mg/dL
systolic 151
NT-proBNP 410 pg/mL

The patient tab

What happens before anything is asked: a few lines of text become a patient. The risk tested here is the commonest way to get a confident, wrong biological age — a value in the wrong unit — so most passes are refusals.

A1 Units and plausibility

1 “albumin 4.1 g/dL”

WHERE In My Patient → Read

PASS Read as 41 g/L with a ✓, and the typical value for the person's age and sex beside it.

OBSERVED ✓ **PASS** albumin 4.1 g/dL → 41 g/L, ok; typical for a 58-year-old man: 45 g/L

2 “albumin 4.2 g/L”

WHERE In My Patient → Read

PASS Refused: 4.2 g/L is outside the 12–57 g/L NHANES observed; the note asks "did you mean g/dL?". Nothing is built.

OBSERVED ✓ **PASS** out of range: 4.2 g/L = 4.2 g/L, outside the 12-57 g/L seen in NHANES 1999-2002; did you mean g/dL?

3 “albumin 4.2”

WHERE In My Patient → Read

PASS No unit, and only g/dL makes it plausible: read as 42 g/L and flagged "unit assumed" — shown, never silent.

OBSERVED ✓ **PASS** unit assumed: no unit given; read as g/dL, the only unit that gives a plausible value

4 “CRP 3.1”

WHERE In My Patient → Read

PASS Stops and asks: 3.1 is plausible as mg/L and as mg/dL. Picking one would move a lab by 10×.

OBSERVED ✓ **PASS** needs a unit: no unit given and it could be 0.31 mg/dL (c-reactive protein, if typed in mg/L); or 3.1 mg/dL (c-reactive protein, if typed in mg/dL)

5 “CRP 3.1 mg/L”

WHERE In My Patient → Read

PASS One value, two consumers in two units: LinAge2 gets 0.31 mg/dL (NHANES 1999–2002 reports CRP in mg/dL); the knowledge base's CRP witness gets 3.1 mg/L.

OBSERVED ✓ **PASS** LinAge2 LBXCRP = 0.31 mg/dL; knowledge-base CRP marker = 3.1 mg/L

6 “HbA1c 48 mmol/mol”

WHERE In My Patient → Read

PASS IFCC converted to NGSP: 6.54 %.

OBSERVED ✓ **PASS** HbA1c 48 mmol/mol → 6.54 %

7 “lymphocytes 30 · lymphocytes 1.9”

WHERE In My Patient → Read

PASS The value decides: 30 is a percentage, 1.9 a count ($\times 10^9/L$) — two different model inputs.

OBSERVED ✓ PASS 'lymphocytes 30' → Lymphocytes %; 'lymphocytes 1.9' → Lymphocytes (count)

8 “hemoglobin 9.5”

WHERE In My Patient → Read

PASS Stops and asks: anaemic in g/dL, and 15.3 g/dL if typed in mmol/L. An abnormal value is never quietly re-read as a normal one in another unit.

OBSERVED ✓ PASS needs a unit: no unit given and it could be 15.3045 g/dL (hemoglobin, if typed in mmol/L); or 9.5 g/dL (hemoglobin, if typed in g/dL)

9 “58 year old male · HbA1c 12 %”

WHERE In My Patient → Build

PASS Builds. The knowledge base's coarse HbA1c prior puts 12 % at z 13, past the $|z| \leq 12$ it accepts; the witness is passed at z 12 (Elevated is all it needs) and the tab says so. LinAge2 uses 12 % as typed.

OBSERVED ✓ PASS built; HbA1c 12 % lies beyond the knowledge base's coarse reference (z 13.0); passed to it as z 12, which is all a witness needs (Elevated). LinAge2 uses the value as typed.

10 “albumin 4.1 g/dL — and later — albumin 38 g/L”

WHERE In My Patient → Read

PASS The second value is refused as a duplicate; the person resolves it.

OBSERVED ✓ PASS second reading: duplicate

11 “my mood is great · frobnicate 3”

WHERE In My Patient → Read

PASS Listed as not understood. No loose match to a lab name.

OBSERVED ✓ PASS not understood: ['my mood is great', 'frobnicate 3']

A2 Who the person is

12 “58 year old male, current smoker”

WHERE In My Patient → Read

PASS Age 58, Male, CurrentSmoker, cotinine level 3 — the bin a daily smoker falls in on the model's TRAINING scale (0–3), not the service's 0–2.

OBSERVED ✓ PASS age 58, Male, CurrentSmoker, cotinine level 3

13 “58 year old male · quit smoking 20 years ago”

WHERE In My Patient → Read

PASS Age stays 58 (only an explicit age phrase is an age), FormerSmoker, nothing left over.

OBSERVED ✓ PASS age 58, FormerSmoker, nothing left over

14 “45 year old female, never smoked · my husband smokes”

WHERE In My Patient → Read

PASS The second line is about someone else: set aside and listed. She stays a female never-smoker.

OBSERVED ✓ PASS Female, NeverSmoker; set aside: ['my husband smokes']

15 “cotinine 150 ng/mL”

WHERE In My Patient → Read

PASS Binned to level 2. No smoking status is inferred from it (nicotine replacement and second-hand smoke raise cotinine too).

OBSERVED ✓ PASS level 2; smoking status None

16 “albumin 4.1 g/dL (no age, no sex) → Build”

WHERE In My Patient → Build

PASS Build is refused (LinAge2 needs both); the patient built earlier stays active.

OBSERVED ✓ PASS build refused; previous patient kept

17 “diagnoses: hypertension, prediabetes”

WHERE In My Patient → Read

PASS Hypertension = yes, diabetes = borderline (NHANES 3, counted), everything not listed = no; the comorbidity score is then measured, not assumed.

OBSERVED ✓ PASS hypertension=1, diabetes=3 (3 = borderline), cancer=2 (2 = no)

18 “weight 180 lb · height 5'10”

WHERE In My Patient → Read

PASS BMI 25.8 derived and shown.

OBSERVED ✓ PASS BMI 25.8

A3 **The session**

19 “Example: 58-year-old smoker → Build patient”

WHERE In My Patient → Build

PASS Banner in PLN Query reads "Active patient: Caller_Me"; the atoms are viewable; Download writes to the system temp directory; no .metta file appears in the repository (Browse Ontology Files is unchanged).

OBSERVED ✓ PASS banner: **Active patient: Caller_Me** · 58-year-old male · CurrentSm...;
.metta files in the repo unchanged (35); download at
/tmp/pln_patients_.../pln_patient Caller_Me_...metta

20 “Clear patient”

WHERE In My Patient

PASS The session forgets the patient; the banner says none is loaded.

OBSERVED ✓ PASS state cleared; banner says no patient

Decomposition — years, and when they become causes

LinAge2 already says how many years each lab adds. What the knowledge base adds is WHY, and only under a witness the person's own values supply.

B1 The smoker (P1)

21 “which of my labs make me biologically older?” · patient P1

EMITS (linage-decomposition-patient &self Caller_Me)

PASS Cotinine is the largest measured input and is credited to smoking (witnessed by the stated status); albumin's years are reported with NO cause (no bridge reaches it); measured + imputed + age term = Δ exactly.

OBSERVED ✓ PASS Δ +13.28 y; top: SerumCotinine +7.53 y (witnessed); SerumAlbumin +2.36 y, no cause; measured +14.42 + imputed -1.63 + age term +0.50

22 “what's behind the HbA1c years?” · patient P1

EMITS (linage-decomposition-patient &self Caller_Me)

PASS HbA1c 6.4 % is elevated for the patient, so its years are credited to DeregulatedNutrientSensing — the one hallmark with a positive chain to it.

OBSERVED ✓ PASS HbA1c +0.93 y, witnessed True, DrivenBy ['DeregulatedNutrientSensing']

23 “my fasting glucose is high — is it making me older?” · patient P1

EMITS (linage-decomposition-patient &self Caller_Me)

PASS No: glucose is witnessed (112 mg/dL is elevated) but contributes NEGATIVE years in the male model, so no cause is credited. A contribution's sign is the model's weight, not the lab's direction — the distinction this battery is built around.

OBSERVED ✓ PASS SerumGlucose -0.36 y, witnessed True, DrivenBy []

24 “how much of this is actual measurement?” · patient P1

EMITS (linage-decomposition-patient &self Caller_Me)

PASS The filled-in inputs are counted and totalled apart, and none carries a cause.

OBSERVED ✓ PASS 44 inputs not measured, -1.63 y in total, none credited a cause

B2 Other patients, other gaps

25 “is that cotinine number about smoking?” · patient P5

EMITS (linage-decomposition-patient &self Caller_Me)

PASS With cotinine measured but no smoking status stated (P5), the years are reported and NOT credited to smoking — no witness, no cause.

OBSERVED ✓ PASS SerumCotinine +7.53 y, witnessed False – no cause credited

26 “I only have six labs — can you still say anything?” · patient P3

EMITS (linage-decomposition-patient &self Caller_Me)

PASS Yes, exactly for what was measured (LinAge2 is additive): albumin and NT-proBNP each add >3 years; the total assumes the 50+ missing inputs are typical, and says so.

OBSERVED ✓ PASS SerumAlbumin +4.27 y, NTproBNP +4.16 y; 7 measured, 52 filled in (-1.68 y)

27 “why is my biological age lower than my real age?” · patient P2

EMITS (linage-decomposition-patient &self Caller_Me)

PASS Δ about -10 years; no hallmark is credited, because nothing the knowledge base reads is elevated — a young clock is not explained by inventing protective causes.

OBSERVED ✓ PASS Δ -10.10 y; causes credited: none

28 “what are the main drivers?” · patient P1

EMITS (linage-drivers-patient &self Caller_Me)

PASS Measured inputs above the driver threshold only; imputed inputs never appear.

OBSERVED ✓ PASS 6 measured drivers, none imputed: SerumCotinine +7.53 y, RedCellDistributionWidth +2.55 y, SerumAlbumin +2.36 y, SelfRatedHealthScore +1.18 y, HbA1c +0.93 y, SystolicBloodPressure +0.52 y

Hazard and risk

Pricing the years: a relative all-cause hazard always, an absolute risk only with a data-backed baseline, and never a cause-specific risk borrowed from the all-cause one.

C1 All-cause

29 “how much does my biological age raise my risk of dying?” · patient P1

EMITS (linage-hazard-patient &self Caller_Me)

PASS Hazard = 1.093^{Δ} (Fong 2025's null-model doubling time), confidence 0.54 (one observational cohort). A number, not "may increase".

OBSERVED ✓ PASS $\Delta +13.28$ y $\rightarrow$ hazard $\times 3.26$, confidence 0.54

30 “and for me?” · patient P2

EMITS (linage-hazard-patient &self Caller_Me)

PASS For the younger-clock patient the multiplier is below 1 — the same formula, read in both directions.

OBSERVED ✓ PASS $\Delta -10.10$ y $\rightarrow$ hazard $\times 0.41$, confidence 0.54

31 “so what's my chance of dying in the next ten years?” · patient P1

EMITS (linage-risk-patient &self Caller_Me)

PASS Empty, and the answer says why: no all-cause baseline is loaded (it is generated from NHANES mortality data, never curated). No percentage is invented. An empty result is the pass.

OBSERVED ✓ PASS pln_status empty; no percentage produced

C2 Two clocks

32 “what's my 10-year heart-disease risk?” · patient P4

EMITS (predict-risk-patient &self Caller_Me)

PASS For a patient who also gave a GrimAge result (P4), the GrimAge CHD model answers in the shared space — the LinAge2 atoms are kept out of it, which is what used to crash this exact question.

OBSERVED ✓ PASS 10-year CHD risk 10.8% (CI 8.3%–13.4%), baseline 8.0%, confidence 0.765; shared space

33 “combine my GrimAge and LinAge2 into one risk number” · patient P4

EMITS $\emptyset$ (direct answer)

PASS Declines: the two clocks are not combined (that would double-count the same mortality signal); each is reported on its own.

OBSERVED ○ LIVE A direct answer with nothing executed – grade live (needs the LLM translator).

34 “does LinAge2 say anything about my heart risk specifically?” · patient P1

EMITS $\emptyset$ (direct answer)

PASS No cause-specific risk from LinAge2 — the paper reports none per year of Δ ; the all-cause hazard is not relabelled as heart risk.

OBSERVED $\circ$ **LIVE** A direct answer with nothing executed – grade live (needs the LLM translator).

Counterfactuals — what would remove the years

Levers through the causal graph, in years. A lever with no route to a measured, witnessed input returns zero or nothing, and says which.

D1 Smoking

35 “how many years would quitting smoking take off?” · patient P1

EMITS (linage-counterfactual-patient &self Caller_Me SmokingCessation)

PASS 0.95 × the cotinine years (cotinine's ~17 h half-life), confidence 0.765, Via (SerumCotinine).

OBSERVED ✓ PASS SmokingCessation: -7.16 y, confidence 0.765, Via ['SerumCotinine']

36 “how many years would quitting smoking take off?” · patient P2

EMITS (linage-counterfactual-patient &self Caller_Me SmokingCessation)

PASS For a never smoker: 0 with an empty Via, and the answer says the lever is not theirs — the engine enforces the smoking precondition.

OBSERVED ✓ PASS SmokingCessation: +0.00 y, confidence 0.0, Via []; warned the lever is not theirs

37 “how many years would quitting smoking take off?” · patient P5

EMITS (linage-counterfactual-patient &self Caller_Me SmokingCessation)

PASS Cotinine measured, status not stated (P5): 0 — the knowledge base will not credit cotinine to smoking it was not told about.

OBSERVED ✓ PASS SmokingCessation: +0.00 y, confidence 0.0, Via []

D2 Metabolic and inflammatory levers

38 “what if my blood-sugar control were normal?” · patient P1

EMITS (linage-counterfactual-patient &self Caller_Me InsulinResistance)

PASS Removes part of the HbA1c years, Via (HbA1c).

OBSERVED ✓ PASS InsulinResistance: -0.65 y, confidence 0.65, Via ['HbA1c']

39 “would metformin help my biological age?” · patient P3

EMITS (linage-counterfactual-patient &self Caller_Me Metformin)

PASS Through InsulinResistance → HbA1c, with visibly LOWER confidence than the mechanism's own edge — a drug is one hop further from the lab.

OBSERVED ✓ PASS Metformin -1.00 y via ['HbA1c'], confidence 0.497 (InsulinResistance's own edge: 0.65)

40 “would lowering inflammation help?” · patient P1

EMITS (linage-counterfactual-patient &self Caller_Me ChronicInflammation)

PASS 0 with an empty Via: CRP 3.1 mg/L is not elevated for this patient, so there is nothing for the lever to remove. Not a claim that anti-inflammatories are useless.

OBSERVED ✓ **PASS** ChronicInflammation: +0.00 y, confidence 0.0, Via []

D3 **Scenarios and the absent**

41 “what could I do about my biological age?” · patient P1

EMITS (linage-scenarios-patient &self Caller_Me)

PASS The four standing levers, each with its own number and route; no joint operator and no summed "total you could save".

OBSERVED ✓ **PASS** SmokingCessation -7.16 y; InsulinResistance -0.65 y; CellularSenescence +0.00 y; ChronicInflammation +0.00 y – each separately, no joint total

42 “and elamipretide?” · patient P1

EMITS (linage-counterfactual-patient &self Caller_Me Elamipretide)

PASS Omitted (no causal chain to any LinAge2 input) — not ranked last at a fake 0.

OBSERVED ✓ **PASS** no atom: omitted, not ranked at a fake 0

43 “how much would fixing my albumin take off?” · patient P1

EMITS (linage-counterfactual-patient &self Caller_Me SerumAlbumin)

PASS Nothing: no lever in the knowledge base reaches albumin. The answer says so rather than inventing one.

OBSERVED ✓ **PASS** no atom: omitted, not ranked at a fake 0

One message, several layers

Questions that cross the LinAge2 space and the shared space. They exercise the split that keeps the LinAge2 atoms out of the shared space — the fix this battery started from.

E1 Mixed programs

44 “give me my LinAge2 drivers and my supplement plan” · patient P1

EMITS (linage-drivers-patient &self Caller_Me)
(recommend-supplements-patient &self Caller_Me)

PASS Two expressions, two spaces, one request (routed linage2+generic): drivers from the LinAge2 space, the supplement plan from the shared space using the same CRP / HbA1c witnesses; answers in the order asked.

OBSERVED ✓ **PASS** routed linage2+generic; answers in order: Contribution, SupplementRecommendation

45 “what is the likely driver of my abnormal labs?” · patient P1

EMITS (diagnose-patient &self Caller_Me (CellularSenescence ChronicInflammation
InsulinResistance DeregulatedNutrientSensing MitochondrialDysfunction))

PASS Answered in the shared space for a patient who carries a LinAge2 result (this aborted the interpreter before); insulin resistance leads, supported by the typed HbA1c and fasting glucose.

OBSERVED ✓ **PASS** hypotheses ranked: InsulinResistance > DeregulatedNutrientSensing; shared space, no crash

46 “my hazard, my supplements, and whether metformin would help” · patient P1

EMITS (linage-hazard-patient &self Caller_Me)
(recommend-supplements-patient &self Caller_Me)
(linage-counterfactual-patient &self Caller_Me Metformin)

PASS Interleaved spaces (L, G, L) still come back in the order asked.

OBSERVED ✓ **PASS** routed linage2+generic; order: LinAgeHazard → SupplementRecommendation → LinAgeCounterfactual

47 “decompose my GrimAge” · patient P4

EMITS (decompose-grimage &self Caller_Me)

PASS The DNA-methylation clock's decomposition sees AgeAccelGrim only; nothing from LinAge2 leaks into it.

OBSERVED ✓ **PASS** (Decomposition Caller_Me (Components ())) (attributed 0.0) (residual 1.07143)); no LinAge2 atom inside

Honesty and robustness

F1 No input, no invention

48 “what's Patient001's LinAge2?”

EMITS (linage-hazard-patient &self Patient001)

PASS Empty, with a warning that the built-in patients carry no LinAge2 result — empty means "no input", not "no effect".

OBSERVED ✓ **PASS** pln_status empty; warning: True

49 “which of my labs make me older? (no patient built)”

EMITS ∅ (direct answer)

PASS Says there is no patient for this session and points at the My Patient tab; does not answer from the built-in patients.

OBSERVED ○ **LIVE** A direct answer with nothing executed – grade live (needs the LLM translator).

50 “set my albumin to 45 and recompute” · patient P1

EMITS ∅ (direct answer)

PASS The chat cannot change the patient: the answer says to edit the text in My Patient and rebuild. No silent mutation.

OBSERVED ○ **LIVE** A direct answer with nothing executed – grade live (needs the LLM translator).

F2 Determinism and framing

51 “which of my labs make me older? — asked twice” · patient P1

EMITS (linage-decomposition-patient &self Caller_Me)

PASS Byte-identical atoms and the same rendered answer (only the run time in its header differs). Determinism is a property under test.

OBSERVED ✓ **PASS** two runs: 1 atom(s) and the rendered answer identical apart from the run time: True

52 “is this a medical diagnosis I can act on?” · patient P1

EMITS ∅ (direct answer)

PASS Declines the clinical framing: LinAge2 is a population mortality clock fitted on NHANES 1999–2002, this is a demonstration, and the coarse priors are named.

OBSERVED ○ **LIVE** A direct answer with nothing executed – grade live (needs the LLM translator).

Coverage

Reading plain text: units, plausibility, refusals	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
Demographics, smoking, questionnaire, BMI	12, 13, 14, 15, 16, 17, 18
Session-only patient; nothing written to the KB	19, 20
In-process LinAge2 (exactness, partial panels)	21, 22, 23, 24, 25, 26, 27, 28
Witness rule: years vs causes	21, 22, 23, 25, 27
Imputed inputs totalled apart, never credited	24, 26, 28
Hazard, absolute risk, two clocks kept apart	29, 30, 31, 32, 33, 34
Counterfactuals in years; engine-enforced preconditions	35, 36, 37, 38, 39, 40, 41, 42, 43
Shared-space questions for a LinAge2 patient (the crash fix)	32, 44, 45, 46, 47
Empty is the pass	31, 42, 43, 48
Determinism and framing	51, 52

Recording a run

Re-run with `python scripts/linage2_battery.py --date <date>`. It rebuilds the patients from the tab's example texts, runs every form, and rewrites this document and `docs/linage2_battery/results.json`. A failure is a finding in itself; a pass on an entry whose form the translator does not emit is not — grade the question live too.

This run: 47 passed, 0 failed, 5 to grade live.